Sakala AI · One Request → Exact Officer → Done On Time

Built and running (14 Oct 2026): a citizen writes, speaks or sends a photo on WhatsApp; Sakala AI understands it, routes it to the exact field officer of the right department down to the Gram Panchayat, keeps the officer on the clock, and gives the Chief Secretary a live view with charts and a Karnataka map.

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Government of Karnataka
Praja Seva Department · Sakala Mission
Live POC · Mysuru pilot · 14 October 2026

Why this is needed (from the real data shared)

- **6 dropdowns** picked **by hand**; wrong picks cause misrouting
- **39%** of Mobile App complaints **stuck in manual scrutiny** before reaching any officer
- **6 depts** get the **same drinking-water problem**
- **349** complaints **parked at one state desk**
- **64 / 3,491** closures have **citizen feedback**
- **1.55M** Sakala applications **delivered late** in 2025 (4.5% of 35.2M)

HOW IT WORKS · 8 STAGES, ONE PIPELINE FOR EVERY CHANNEL (AS BUILT)

1 Entry Channels
How does a request come in?

- **WhatsApp** (no app to install): text, voice note, photo of a petition, live location, in Kannada / English
- Two-way: officer questions, decisions and resolutions arrive back in the same chat; the citizen replies, confirms 👍 / 👎 or appeals
- Same pipeline for the 1902 call centre, Praja Seva camps, Grama One / CSC counters and the app
- Excel / CSV upload for departments with spreadsheets; officer web console

Solves UC 1 · 2 · 23

2 Intake & Privacy
What did the citizen say?

- Speech-to-text (Kannada / English) on the government's own model gateway
- Vision model reads the petition photo; unreadable or unrelated images are refused, never guessed
- Language detection + one-line English summary
- **Masks** phone, Aadhaar, email and card numbers before any AI step
- Pulls out Sakala GSC numbers and links them to the application

Solves UC 2 · 4

3 Validate & Complete
Is it complete and clean?

- Village / GP resolved from the message (Kannada names transliterated), GPS, or a pick-list; taluk-level services skip the village question
- Duplicate check: same service, same GP → linked, not re-filed
- Per-service checklist for Sakala applications (name, village, document photo)
- Wrong-door check: RTI, other departments → told where to go

↺ AI agent keeps asking the citizen until the application is complete. Sakala's legal clock starts only then.

Solves UC 1 · 3 · 5

4 AI Understanding
WHAT is the problem?

- Finds the 5 most similar real grievances (9,999 from 3–9 Oct 2026)
- The AI model picks **Department → Line Dept → Service → Grievance** from the official master only (enum-constrained)
- A second, offline model trained on the real data must agree; keyword rules override for RTI / other departments / new applications
- Confidence below a threshold tuned from measured accuracy → scrutiny desk with the top-3 options
- Vulnerability flag: elderly, widow, disabled, pension, ration

Solves UC 6 · 9

5 Route to Officer
WHO owns it, here?

- Service → officer role learned from the real assignments (Khata → Tahsildar 100%, water → AE RDWS 100%, street light → PDO 97%)
- Role + taluk + GP → one named officer: PDO of that GP, Tahsildar / ADLR / Food Inspector / AEE of that taluk
- **97.6%** agreement with the real iPGRS officer on 673 Mysuru cases; 92.5% end to end from text alone
- Every decision stores its evidence: confidence, threshold, rules, similar cases, location source, officer rule

⚑ Unsure, outside the master, or village unknown? → Scrutiny desk (L2/L3) decides and re-routes; the citizen is told. Never a silent guess.

Solves UC 7 · 8 · 10

6 Deadlines & Alerts
Is the officer acting on time?

- Grievance deadlines per level (7 / 5 / 3 days, assumed) + the Sakala legal clock for applications
- Due-tomorrow reminders, then **automatic 3-level escalation**; the citizen is told each time
- Officer inbox: open cases, days left, vulnerable first, accept / ask info / return / reject (reason mandatory) / resolve
- ₹20/day (max ₹500) ordered by the Competent Officer → Form C notice → **HRMS debit** (mock)
- Officer messages translated to Kannada for Kannada chats

Solves UC 11 · 12 · 13 · 14

7 Resolution Check
Was it really solved?

- AI compares the complaint with the officer's Action Taken Report
- Score 5 Fully resolved … 1 Irrelevant; below 4 the case stays open and is flagged
- AI drafts a first ATR the officer edits
- Citizen asked on WhatsApp "was it solved?" · 👎 re-opens the case and informs the senior

↺ Not resolved, or the citizen says no? → re-opened, back to stage 6 with the senior officer informed.

Solves UC 15

8 Leadership View
What should leaders act on?

- Command dashboard with a live **Karnataka map**: districts shaded by real 2025 late delivery; zoom to taluk; on-time / overdue / vulnerable filters
- **Sakala district tracker**: real 2025 reports, early warning for slipping districts, fairer scorecard
- Insights: hotspots with AI-suggested fixes, Praja Seva history, misrouting detector, data quality
- Wednesday Review Pack (EN + ಕನ್ನಡ) for Tahsildar/EO → AC → DC → CS; Form E-4 list
- **AI mode**: ask anything, answered only from data tools, with bar / line / donut charts

⚠ Accuracy proof screen: live test on real complaints (dept 97.5% · service 95% · officer 92.5% on 40 cases).

Solves UC 16 – 22

A · One complaint, end to end (as the demo runs it)

"ನಮ್ಮ ಗೊರಹಳ್ಳಿ ಗ್ರಾಮದಲ್ಲಿ ಕಳೆದ ಒಂದು ವಾರದಿಂದ ಕುಡಿಯುವ ನೀರು ಬರುತ್ತಿಲ್ಲ."

"No drinking water in our Gorahalli village for the last one week."

Source: WhatsApp (text) · village read from the Kannada message → Chapparadahalli GP, Piriyapatna

Today: 6 dropdowns picked by an operator; 39% of app complaints wait in manual scrutiny.

Department	RDPR
Line Dept	Rural Drinking Water & Sanitation
Service	Provision of Drinking Water
Confidence	95% · offline model agrees · 4 of 5 similar real cases
Location	Gorahalli → Chapparadahalli GP → Piriyapatna
Officer (L1)	AE/JE Rural Drinking Water, Piriyapatna (100% of 229 real cases)
Escalates to	EO TP Piriyapatna → ZP CEO Mysuru

Routed in about 3 s; the citizen gets the officer's name and deadline on WhatsApp.

Grievance timelines (7 / 5 / 3 days) and Sakala service timelines are assumed until confirmed by the department.

B · Two escalation ladders, never mixed

Grievance (complaint) · iPGRS / Janaspandana

LMF (L1) field officer → L2 next senior → L3 top level

Moves up automatically when the deadline passes; the citizen can escalate twice.

Sakala service application · GSC number

Designated Officer delivers service → Competent Officer 1st appeal · 30 days → Appellate Authority 2nd appeal · 60 days

Late service: ₹20/day up to ₹500, recovered from the officer's salary via HRMS.

Sakala compensation applies only to Sakala applications, never to ordinary complaints.

C · Foundation: masters, data & connectors

Official masters (single source of truth)

6-level category master · Jurisdiction & officer master · Sakala service timelines (assumed) · Holiday calendar · Full LGD village master

Data used in the POC

9,999 real grievances (masked) · Sakala district reports 2025 (35.2M applications) · 701 real Mysuru cases with their real officers · 2 Sakala applications + 10 grievances for the demo · Sample scanned forms

Connectors for every digital level

Excel / CSV upload (e.g. monthly reports) · Paper scan · Officer console · Department APIs (mock) · HRMS (mock)

Stack: Python (FastAPI) · SQLite · government model gateway (text + vision model, Kannada speech-to-text) inside the network · offline fallback model trained on the real data · single-page web app with Chart.js and a KGIS/OSM map.

D · Guardrails & what comes next

- AI chooses **only from official masters**; anything else goes to a person
- Auto-route threshold **tuned from measured accuracy**; below it a human decides with top-3 options
- Every routing decision keeps its **evidence** (confidence, rules, similar cases, officer rule) for audit
- AI mode answers **only from data tools**; instruction-override and off-data prompts are refused and logged
- Unreadable photos / silent audio are refused, never guessed; every AI call is logged for weekly review
- A chat sees only its own cases; only the owner acts, only the Competent Officer compensates; ladders never mix
- Rejection needs a reason and the citizen is told the appeal officer (reply APPEAL); 👎 re-opens
- Per-client rate limits and size caps, as in the sister POCs; all rendered text sanitised

Production phase (shown in UI only)

AI callback to citizen · Namma Seva ★ rating · Live iPGRS / Sakala / HRMS · SMS / WhatsApp alerts · Learning loop · Early-warning alerts

Right department · Exact officer, down to village / GP · Nothing silently delayed · Accountability by law · Leaders see everything